

Seed 1, Layer 3 (Round 2): a Pascal ladder for A_2 fermionic forms and the q -binomial transform reformulation of the Monotonicity Conjecture

proof-search agent, seed 1

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Setting. Fix rank 3, level d with $\gcd(d, 3) = 1$, and for a profile $c = (c_0, c_1, c_2)$, $c_0 + c_1 + c_2 = d$, let $F_{c,m}(q)$ be the generating function of cylindric partitions of profile c with largest part at most m , and $H_{c,m} := (q; q)_m F_{c,m}$. The $H_{c,m}$ are polynomials with nonnegative coefficients, constant on ρ -orbits of profiles, and satisfy the level tower $(1 + q^m + q^{2m}) H_{c,m} = \sum_{c'} q^{m \cdot \text{EMD}(c, c')} H_{c', m-1}$ (target-first orientation, validated against brute-force enumeration in the session engine). The *Monotonicity Conjecture* (the layer-3 bottleneck) states $H_{c,m} \geq H_{c, m-1}$ coefficientwise. All inequalities between polynomials below are coefficientwise. $\begin{bmatrix} m \\ n \end{bmatrix}$ denotes the Gaussian binomial, $\begin{bmatrix} m \\ n \end{bmatrix} = 0$ unless $0 \leq n \leq m$.

1 The one-binomial lemma

Lemma 1 (one-binomial monotonicity lemma). *Let $(F_m)_{m \geq 0}$ be given by $F_m = \sum_{n \geq 0} \varphi_n(q) \begin{bmatrix} m \\ n \end{bmatrix}$ where the $\varphi_n \in \mathbb{Z}[q]$ (or Laurent) do not depend on m and have nonnegative coefficients. Then*

$$F_m - F_{m-1} = \sum_{n \geq 1} \varphi_n(q) q^{m-n} \begin{bmatrix} m-1 \\ n-1 \end{bmatrix} \geq 0.$$

Proof. q -Pascal: $\begin{bmatrix} m \\ n \end{bmatrix} - \begin{bmatrix} m-1 \\ n \end{bmatrix} = q^{m-n} \begin{bmatrix} m-1 \\ n-1 \end{bmatrix}$. □

Every sequence of polynomials admits exactly one such expansion:

Lemma 2 (inverse q -binomial transform). *For any sequence $(H_m)_{m \geq 0}$ in $\mathbb{Z}[q]$ there are unique $g_n \in \mathbb{Z}[q]$ with $H_m = \sum_{n=0}^m g_n \begin{bmatrix} m \\ n \end{bmatrix}$ for all m ; namely $g_m = H_m - \sum_{n < m} g_n \begin{bmatrix} m \\ n \end{bmatrix}$, equivalently $g_n = \sum_{k=0}^n (-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix} H_k$ (g_n is an n -th q -difference of the sequence).*

Conjecture 3 (g -positivity). *For every level d with $\gcd(d, 3) = 1$ and every profile orbit c , the inverse q -binomial transform $g_{c,n}$ of $(H_{c,m})_m$ has nonnegative coefficients.*

By Lemma ??, **Conjecture ?? implies the Monotonicity Conjecture**, with the exact surplus $H_{c,m} - H_{c, m-1} = \sum_{n \geq 1} g_{c,n} q^{m-n} \begin{bmatrix} m-1 \\ n-1 \end{bmatrix}$. The statement of Conjecture ?? contains *no* bound variable m : one fixed polynomial sequence per orbit. Verified exactly (integer arithmetic) for all orbits at $d = 2, 4, 5$ for $n \leq 8$ and $d = 7$ for $n \leq 6$ (script `seed1_R2L3_qbt.py`). At $d = 2$ it is a theorem: $g_n = q^{n^2}$ (balanced), q^{n^2+n} (corner). The valuation data $\text{val } g_{c,n}$ stabilizes in d for corresponding profile shapes (e.g. $(0, 0, d)$: $0, 2, 6, 11, 18, 26, 36, \dots$ for all of $d = 4, 5, 7$), suggesting a uniform fermionic description of $g_{c,n}$.

2 The A_2 Pascal ladder

For $a, b, c \in \mathbb{Z}$ define the Warnaar-type bounded fermionic form

$$\Phi_m(a, b, c) := \sum_{n, j \geq 0} q^{n^2 - nj + j^2 + an + bj} \begin{bmatrix} m \\ n \end{bmatrix} \begin{bmatrix} 2n + c \\ j \end{bmatrix}.$$

$\Phi_m(a, b, c)$ has nonnegative coefficients for all a, b, c (each summand is a monomial times Gaussian binomials).

Theorem 4 (A_2 ladder identity). *For all $a, b, c \in \mathbb{Z}$ and $m \geq 1$,*

$$\Phi_m(a, b, c) - \Phi_{m-1}(a, b, c) = q^{m+a} \Phi_{m-1}(a+1, b-1, c+2).$$

In particular $\Phi_m(a, b, c)$ is coefficientwise monotone in m , as are all its iterated surpluses, which move along the lattice ray $(a, b, c) \mapsto (a+1, b-1, c+2)$.

Proof. Only the first binomial depends on m . By q -Pascal, $\begin{bmatrix} m \\ n \end{bmatrix} - \begin{bmatrix} m-1 \\ n \end{bmatrix} = q^{m-n} \begin{bmatrix} m-1 \\ n-1 \end{bmatrix}$, so the difference equals $\sum_{n \geq 1, j \geq 0} q^{n^2 - nj + j^2 + an + bj + m - n} \begin{bmatrix} m-1 \\ n-1 \end{bmatrix} \begin{bmatrix} 2n+c \\ j \end{bmatrix}$. Substituting $n \mapsto n+1$, the exponent becomes $n^2 - nj + j^2 + (a+1)n + (b-1)j + m + a$ and the binomials become $\begin{bmatrix} m-1 \\ n \end{bmatrix} \begin{bmatrix} 2n+(c+2) \\ j \end{bmatrix}$. $\square$

Theorem ?? is the exact A_2 analogue of the $d=2$ ladder $E_{a,m} - E_{a,m-1} = q^{m+a} E_{a+1,m-1}$ for $E_{a,m} = \sum_j q^{j^2 + aj} \begin{bmatrix} m \\ j \end{bmatrix}$: the same one-line q -Pascal argument, with the j -binomial $\begin{bmatrix} 2n+c \\ j \end{bmatrix}$ inert. It was verified exactly on the grid $a \in [-2, 3]$, $b \in [-4, 3]$, $c \in [-1, 3]$, $m \leq 7$ (zero failures), and discovered from data: the empirical surplus chain of the $d=4$ tower matches $(1, 1, 0) \rightarrow (2, 0, 2) \rightarrow (3, -1, 4)$ and $(0, 1, 0) \rightarrow (1, 0, 2) \rightarrow (2, -1, 4)$ with prefactor q^{m+a} throughout.

Corollary 5. *The following identities were verified exactly for $m \leq 8$ (previously $m \leq 5$, Layer 2 seed 4): $H_{(1,1,2),m} = \Phi_m(0, 0, 0)$, $H_{(0,0,4),m} = \Phi_m(1, 1, 0)$, $H_{(0,3,1),m} = \Phi_m(0, 1, 0)$ at $d=4$. Conditional on these (the bounded A_2 Andrews–Gordon forms), Theorem ?? proves monotonicity for these three orbits, with explicit surplus $H_m - H_{m-1} = q^{m+a} \Phi_{m-1}(a+1, b-1, c+2)$. The valuations $m+a$ match the tower census (corner orbit $(0, 0, 4)$: $a=1$, extra factor q). Equivalently, in the language of Conjecture ??, these orbits have $g_{c,n} = q^{n^2+an} \sum_j q^{j^2 - nj + bj} \begin{bmatrix} 2n+c \\ j \end{bmatrix}$, manifestly nonnegative.*

3 Negative results (search exclusions)

Exact searches at $m \leq 7$ (scripts `seed1_R2L3_ansatz.py`, `seed1_R2L3_twoterm.py`) exclude, for the two remaining $d=4$ orbits $(0, 1, 3)$, $(0, 2, 2)$ and for all seven $d=5$ orbits:

- any form $\sum_{n,j} q^{An^2+Bnj+Cj^2+an+bj} \begin{bmatrix} m \\ n \end{bmatrix} \begin{bmatrix} \alpha n + \beta m + c \\ j \end{bmatrix}$ with $A \in \{1, 2\}$, $B \in [-2, 1]$, $C \in [1, 3]$, $a, b \in [-2, 4]$, $\alpha \in [0, 3]$, $\beta \in \{0, 1\}$, $c \in [-2, 4]$;
- any two-term decomposition $H = \Phi(p_1) + q^t \Phi(p_2)$ with p_i in the grid $a \in [-1, 4]$, $b \in [-2, 4]$, $c \in [-2, 5]$.

The shift patterns of their iterated surpluses (increments $0, 1, 1, 0, \dots$ rather than the ladder's $+1$ per step in a) show their m -dependence cannot sit in a single $\begin{bmatrix} m \\ n \end{bmatrix}$ of these shapes; a three-variable form (cf. Warnaar's modulus-8 conjecture, variables n_1, n_2, m_1 with $m_2 = 2n_2$) is the natural next ansatz. Note that Conjecture ?? sidesteps this entirely: the $g_{c,n}$ of the missing orbits are nonnegative in all computed cases even though no closed form is known.